HealthConnect Clinic

Week 4 Project Summary — Data Analytics Track

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This summary is deliberately concise and complements — rather than repeats — the full Initial Analysis Document (docs/HealthConnect_Week4_Analysis_Document.docx) and the executed EDA notebook (notebooks/HealthConnect_Week4_EDA.ipynb).

1. The Problem This Track Addresses

As the Data Analytics track, our job this week was not to solve the no-show problem, but to establish whether the data can support solving it: understand what the appointment data contains, check it's trustworthy, and identify which factors are worth pursuing as drivers of missed appointments — output the wider HealthConnect project (Data Science, ML Engineering, Generative AI tracks) can build on.

2. Resources Used

Resource	Use
HealthConnect_Appointment_Data.csv	5,000-row appointment dataset — primary analysis input
HealthConnect_Data_Dictionary.xlsx	Field definitions — reviewed before touching the data
HealthConnect_Clinic_Knowledge_Base.docx	Reviewed for context only; not used in this track's analysis (belongs to the Generative AI track)

3. Key Observations From Week 4

- Data quality is high: no duplicate records, and every internal consistency rule tested (derived-field checks, business-rule checks) passed with zero violations.
- Missingness is limited and mostly explainable: reminder_channel is blank exactly when no reminder was sent (not an error); distance_to_clinic_km and waiting_time_minutes have small (<2%) genuine gaps.
- Two variables show a clearly stronger association with appointment_outcome than the rest: booking lead time and prior no-show history. Reminder usage shows a moderate association. Appointment type, age group, time of day and gender show weak association on their own.
- The dataset supports the Week 4 goal well: there is enough structure and signal to justify five specific business questions and five KPIs for the next stage, without needing additional data collection.

4. Proposed Approach

- Move from associative, descriptive findings (Week 4) to significance-tested findings in the next analytical pass — before treating any variable as a confirmed driver.
- Keep KPI definition and calculation as separate steps, as scoped by the brief: KPIs are identified and justified now, calculated later once the analysis approach is validated.
- Keep this track's output data-only and administrative in framing — no clinical interpretation — consistent with the safety boundaries the Generative AI track's Knowledge Base sets for the parallel chatbot workstream.

5. Key Considerations Affecting the Project

- Two fields need a documented missing-data handling rule before any calculation or modelling uses them (distance_to_clinic_km, waiting_time_minutes).
- 737 records fall on a Sunday; internally consistent within the file, but should be confirmed against real clinic operating-hours assumptions before it's used in any hours-based segmentation.
- Findings so far are associative, not causal — a real risk if they get treated as final conclusions before significance testing.

6. Proposed Focus for Week 5

- Statistical significance testing (chi-square / logistic regression) on the strong and moderate-signal variables identified this week.
- Interaction analysis — e.g. does the lead-time effect hold within each prior-no-show band?
- First KPI calculation pass, now that identification and justification are complete.
- Coordinate with the Data Science track on shared variable definitions, since the same dataset will likely feed a no-show prediction model.

Full documentation: docs/HealthConnect_Week4_Analysis_Document.docx | Code: notebooks/HealthConnect_Week4_EDA.ipynb | Navigation: README.md